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REGULAR ARTICLE

Photophysical properties prediction of selenium- and tellurium-substituted thymidine as potential UVA chemotherapeutic agents

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Abstract Density functional theory and time-dependent density functional theory calculation for a series of photophysical properties (absorption spectra, singlet and triplet excitation energies and spin–orbit matrix elements) have been performed on the sulfur-, selenium- and tellurium-substituted thymine. The heavy atoms have been substituted in 2 or 4 and in both 2,4 position of the thymine ring. Different pathways for the population of the lowest triplet state have been considered. We find that all the considered systems are potential UVA chemotherapeutic agents since the lowest triplet states lie above the energy required for the production of the highly cytotoxic $^1\Delta_g$ excited oxygen molecule and due to the possible and efficient intersystem crossings ensured by high spin–orbit coupling values.

Keywords TDDFT · Photodynamic therapy · Excited states · Substituted thymidines

1 Introduction

Photodynamic therapy (PDT) is a promising and less invasive method for treating a variety of diseases by using visible light irradiation in conjunction with a photosensitizer (PS) that must be non-toxic in the dark [1]. One of the advantages of PDT is its capability to selectively act on the damaged tissue preserving the surrounding normal tissues. Starting from the preparation of Photofrin® (a mixture of dimers and oligomers of hematoporphyrin in which porphyrin units are linked by ether, ester and C–C bonds) in 1983 [2], a series of first- and second-generation PSs have been synthesized or extracted from plants [3–6], tested in vitro and in vivo.

PDT PSs must meet several criteria, and the more important properties are: (1) chemical stability; (2) water solubility; (3) high quantum yield of $^1O_2$ ($^1\Delta_g$) generation; (4) no cytotoxicity in the dark; (5) selectivity and rapid accumulation in target tissues; (6) rapid clearance from patients; (7) a high molar absorption coefficient (ε) in the long wavelength (600–800 nm) that can penetrate deeper tissues.

Although a large number of potential candidates are proposed, very few have shown ideal properties and have been approved for their clinical use (see Ref. [6] and reference therein). For these reasons, recent studies have focused on the development and efficacy of new PSs.

The two most important physical aspects of PDT can be summarized as the processes of light absorption and the energy transfer. The absorption of light in the form of photons induces a transition from the PS ground state S_0 to the excited one S_n with a short lived and loses its energy by emitting fluorescent light or by internal conversion (IC) into heat. Alternatively, it may also undergo a process known as intersystem crossing (ISC) whereby the spin of the excited electron inverts to form a relatively long-lived excited triplet state T_1 [7, 8]. This PS excited triplet state can transfer its energy directly to molecular oxygen that is, consequently excited from the triplet ground state ($^3\Sigma_g$) to the singlet excited state ($^1\Delta_g$), highly reactive and cytotoxic. Alternatively, the excited triplet state can react with a substrate in the environment producing reactive anion and cation radical species that, in turn, may further react with oxygen to produce reactive oxygen species (ROS) that react with the

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tissue. The production of excited singlet oxygen is preferred in PDT and is possible only if the T_1 state of the PS has an energy higher of 0.98 eV necessary to excite the oxygen molecule [9].

Due to their photophysical and photochemical properties and to their possible use in PDT, the thio-substituted nucleobases have been the subject of many experimental and theoretical investigations [10–16]. Special emphasis has been reserved to thymidine since the substitution of oxygen in position 4 or in both 2 and 4 strongly enhances the production of the cytotoxic singlet excited oxygen [10–13]. Although these systems show absorption band that falls down to the so-called therapeutic window (about 600–800 nm) and, consequently, possess a limited light penetration power, they remain good candidates for PDT antimicrobial and antivirus disinfection and UVA chemotherapeutic agents. The available experimental characterization for 2-thiothymine [10], 4-thiothymidine [11, 12] and 2,4-dithiothymidine [13] and the recent accurate theoretical studies on 4-thiothymidine [14] and 2-thiothymine [15] have stimulated our attention to consider the selenium (2-SeT, 4-SeT, 2,4-DiSeT) and tellurium (2-TeT, 4-TeT, 2,4-DiTeT) thymidines also in view of the fact that Se- and Te-containing systems have been previously indicated as potential PSs in PDT since the presence of these heavy atoms redshifts the absorption band and enhances the intersystem crossing [17–19]. In order to assets our computational protocol, we have performed the computations for the thiothymidine derivatives (2-ThioT, 4-ThioT, 2,4-DiThioT) for which a comparison with experimental data and previous theoretical results is possible.

2 Computational details

Our investigation has been performed by using DFT and TDDFT levels of theory previously and extensively tested as reliable tools for the determination of excited-state properties [20–23]. All computations have been performed employing the Gaussian 09 code [24]. Molecular geometries were optimized using the meta-GGA functional M06 [25] coupled with 6-31G* basis set, whereas for the excited-state properties M06/6-31+G* time TDDFT approach has been adopted. This protocol was previously used in a series of systems and properties including the simulation of the photophysical properties of different dye molecules [26–28]. The impact of solvation effects (water) on the considered properties has been estimated by means of the integral equation formalism polarizable continuum model (IEFPCM), which corresponds to a linear response in non-equilibrium solvation [29]. Spin–orbit matrix elements have been computed by using the atomic-mean field approximation [30] as implemented in DALTON code [31], since previous studies have demonstrated as Breit–Pauli method [32] and atomic-mean field approximation provide very similar values [26], the latter of which requires a lower computational cost. Because of the presence of only few hybrid functionals available in DALTON to perform these calculations, B3LYP [33, 34] has been selected and used on the previously M06-optimized geometries, in conjunction with the cc-pVDZ basis set for all the atoms.

3 Results and discussion

The computed electronic transitions are reported in Fig. 1. As a general feature, we observe that the substitution of oxygen atom with sulfur, selenium and tellurium atoms shifts the absorption spectrum to the red. This shift is more pronounced when the substitution concerns the position 4 and increases with the atomic size of the substituent (0.73, 1.02, 1.16 and 1.42 Å for oxygen, sulfur, selenium and tellurium, respectively). This behavior is due to the weakness of the C=X strength in going from C=O to C=S, C=Se and C=Te bonds. Furthermore, the increases of the metallic character and the electron delocalizability along the chalcogens family contribute to the increase in the observed redshift spectral lines. In the case of the 2,4 double substitution, the observed redshift does not correspond to a simple linear combination of the shifts observed in the corresponding 2 and 4 substitution, but is more similar to that occurring when the chalcogens are present in position 4. In all the 2 and 4 substitutions, the transitions are mainly HOMO-1 → LUMO in nature with contribution of HOMO → LUMO ones, while for 2,4 species the contamination coming from other excitations (e.g., HOMO-2 → L, HOMO-3 → L, HOMO → LUMO + 1) appears. From the molecular orbital picture reported in Fig. 2, it is clear that in all the considered systems the S_1 transition and S_2 transition have $n\pi^*$ and $\pi\pi^*$ nature, respectively. The computed absorption maxima well agree with the experimental results available for thymidine (264 nm) [16], 2-thiothymidine (278 nm) [10, 13], 4-thiothymidine (335 nm) [13, 35], 2,4-dithiothymidine (278 and 360 nm) [13] and 4-selenothymidine 369 nm) [36].

As previously mentioned, in a series of recent papers the thioderivatives of thymine and thymidine have been the subject of accurate experimental and theoretical analysis in order to establish their possible use as PS in PDT [10–16]. At this purpose, the excited singlet and triplet states, the spin–orbit matrix elements and the singlet oxygen production upon excitation have been computed or measured. Computational studies have been performed by employing DFT as well as CASCF and CASPT2 levels of theory [14, 15]. In the work of Cui and Fang [15] on the 2-thiothymine, it is shown as DFT, especially when M06-like

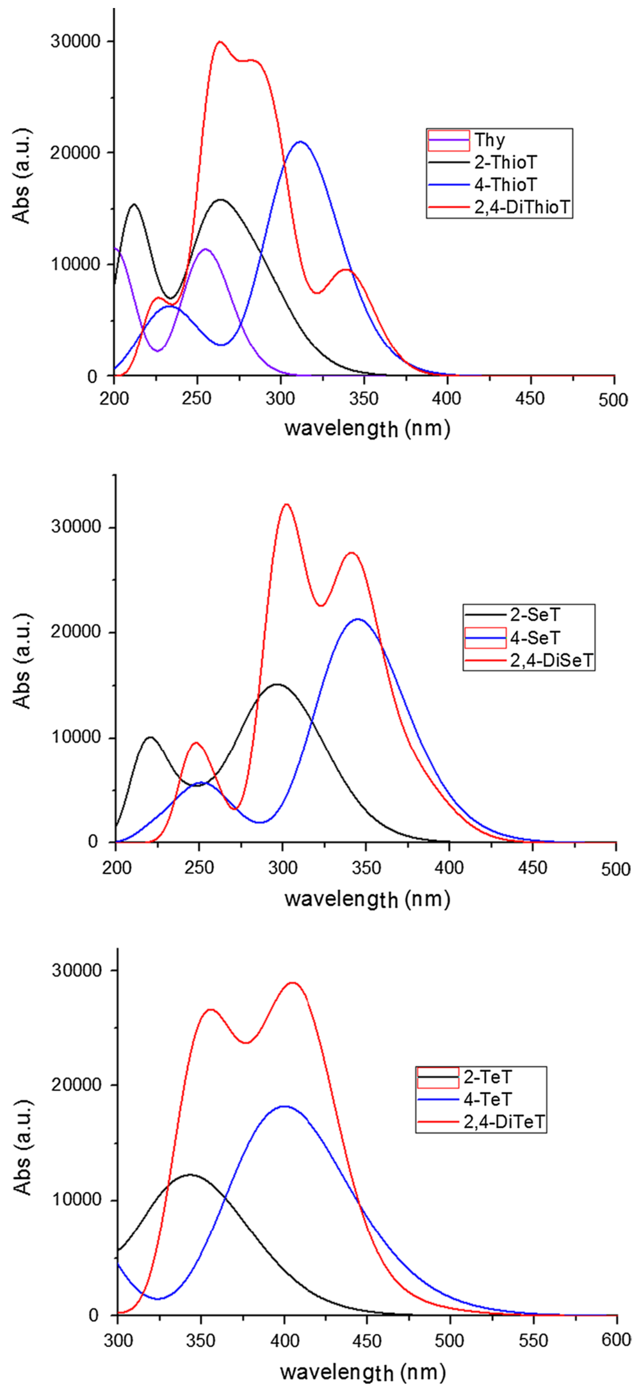

Fig. 1 M06-simulated electronic spectra in water solvent for the studied compounds

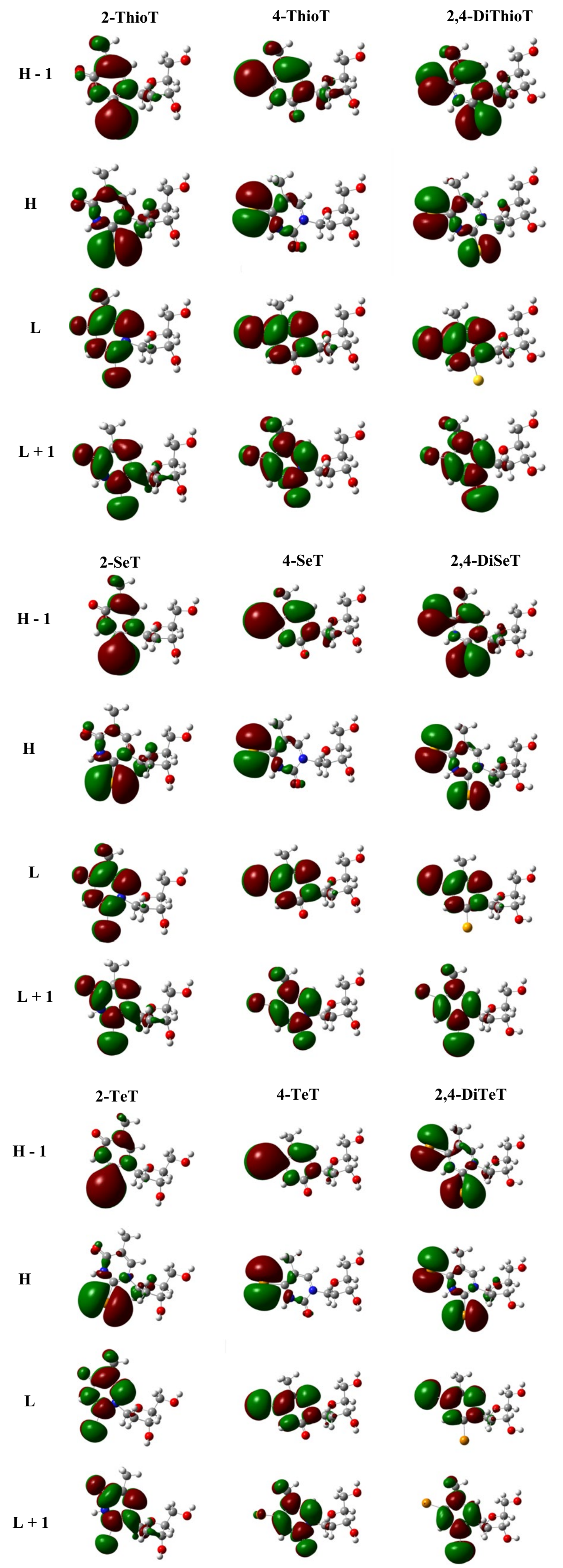

Fig. 2 Orbital composition for the ground state of the studied compounds

exchange–correlation functional is employed, gives singlet and excited states energetic close to the corresponding CASPT2 values. On the other hand, in a series of previous paper our group demonstrated the reliability of DFT in the computations of the spin–orbit matrix elements in a series of PSs active in PDT [23, 37, 38]. The computed main vertical singlet and triplet electronic states and the spin–orbit matrix elements for all the considered systems are reported in Table 1. For thymidine, the spectroscopically bright state is the S_1 ($n\pi^*$), while for all the considered derivatives is the S_2 with a $\pi\pi^*$ character as previously demonstrated in the experimental and theoretical works on the thiothymine and thiothymidines [10–16]. From Table 1, it emerges that all the investigated molecules have a first excited triplet state (T_1) that is higher than the energy required activating the $^1\Delta_g$ excited state of oxygen molecules that is the key cytotoxic agent in PDT. We underline as our computed vertical T_1 value for 2-ThioT is in good agreement with the available experimental counterpart (4.30 vs. 4.14 eV) [15]. Good agreement is also found for the vertical values for S_1, S_2, T_1 and T_2 states computed at CASPT2 level of theory [15]. The activation of singlet excited O_2 is possible if the energy is transferred by the singlet excited state to the triplet one via an intersystem spin crossing that will depend from the values of the spin–orbit matrix elements ($|\hat{H}_{SO}|$) involved in a given transition. As previously suggested in the theoretical study on 4-thiothymidine [11, 14] and on 2-thiothymine [10, 15] and in the experimental investigation on 2-thymine [10], the following most probable pathways for the activation of the lowest triplet state (T_1) are possible: (1) pathI: $S_2 \rightarrow S_1 \rightarrow T_1$; (2) pathII: $S_2 \rightarrow T_2 \rightarrow T_1$. In the first one, the S_2 undergoes a fast IC to S_1 via minimum energy conical insertion S_2S_1. This transition is allowed since the small energy difference between the two singlet excited states. Then the system decays to the T_1 state throughout a ISC process. In the second pathway, the first ISC between the S_2 and T_2 states is followed by an IC from T_2 to T_1. In order to predict the more efficient ISC processes, it is mandatory to compute the spin–orbit matrix elements. As a general rule, the presence of a heavy atom increases the ISC probability via the so-called heavy atom effect. From Table 1, it is evident as the increase in the atom size contributes to significantly increase the $|\hat{H}_{SO}|$. According to the El-Sayed rules [39, 40], the rate of the ISC sensibly increases if the radiationless transition involves a change of orbital type. Therefore, the magnitude of the $|\hat{H}_{SO}|$ between the initial $\Psi(S_n)$ and final $\Psi(T_n)$ wave functions not only depends on the effective nuclear mass of heavy atoms eventually present in the molecular system, but also depends on the nature of the orbitals involved in the coupling. Due to the $\pi\pi^*$ and $n\pi^*$ nature of S_2 and T_1 and S_1 and T_2 states, respectively, we expect high $|\hat{H}_{SO}|$ values for the S_1T_1 and S_2T_2 couplings. Table 1 shows as all the spin–orbit coupling matrix elements increase in going from thymidine (Thy) to the substituted species with high effective nuclear mass. Furthermore, it is evident that the substitution on the position 4 is preferred over the position 2. As expected from the El-Sayed rule and from the lower energetics the $S_1T_1|\hat{H}_{SO}|$ values result to be very high than the corresponding S_2T_2 ones. In any case, from the computed properties, both the investigated paths show efficient intersystem crossing.

We have explored also the possibility that, after the efficient S_2S_1 IC, S_1 can transfer the energy to the T_2 state that in turn can generate singlet oxygen throughout a fast T_2T_1 IC. The $|\hat{H}_{SO}|$ values for this ISC (see Table 1) suggest that, contrary to the previously discussed ISC, the substitution of the heavy atoms in position 2 is more efficient than that in 4. This alternative route is less probable since the smallest spin–orbit coupling obtained. A deep examination of the considered pathway for the formation of the lowest triple state (T_1) shows that only the path I follows the Kasha's rule [41] that indicate as, in a given spin manifold, the transition $S_n \rightarrow S_1 (n > 1)$ should be very fast and triplet-state population (as well as other photophysical properties) should start from the lowest singlet exited state (S_1).

4 Conclusions

The possibility that the sulfur, selenium and tellurium derivatives of thymidine can be proposed as PSs in PDT have been explored theoretically employing density functional theory. The computed singlets and triplets excitations

Table 1 Vertical singlet and triplet energies computed in water at M06/6-31+G* level of theory and spin–orbit matrix elements (cm^{-1}) calculated at the B3LYP/6-31G(d)//M06

	T_1 (eV)	T_2 (eV)	S_1 (eV)	S_2 (eV)	$\lvert\langle\Psi_{S_1}\lvert\hat{H}_{so}\rvert\Psi_{T_1}\rangle\rvert$	$\lvert\langle\Psi_{S_1}\lvert\hat{H}_{so}\rvert\Psi_{T_2}\rangle\rvert$	$\lvert\langle\Psi_{S_2}\lvert\hat{H}_{so}\rvert\Psi_{T_2}\rangle\rvert$
Thy	3.32	4.78	4.87	5.04	31.40	1.13	11.08
2-ThioT	3.18	3.70	3.81	4.30	133.91	25.86	3.53
4-ThioT	2.68	2.94	3.08	3.98	176.02	2.91	64.73
2,4-DiThioT	2.66	2.89	3.03	3.65	172.11	11.62	19.09
2-SeT	2.91	3.28	3.43	3.97	744.56	127.00	22.49
4-SeT	2.20	2.54	2.75	3.60	891.49	17.31	303.67
2,4-DiSeT	2.13	2.47	2.68	3.26	877.92	67.95	74.41
2-TeT	2.67	2.82	2.92	3.45	4716.5	283.13	438.37
4-TeT	1.99	2.13	2.25	3.10	5668.7	35.87	521.43
2,4-DiTeT	1.88	2.01	2.15	2.62	1256.3	159.11	336.73

and the spin–orbit coupling matrix elements and the consideration of different pathways for the production of cytotoxic singlet molecular oxygen allow us to outline the following conclusions:

- absorption energies are short enough to penetrate into the tissue. Consequently, these systems cannot be proposed in the therapy of a series of tumors, but should be used in killing microbes and virus as well in other diseases in which a deeper penetration into the tissue is not requested;
- the lowest triplet excited state (T_1) lies up to the energy requested for the $^1\Delta_g$ singlet oxygen production in all the considered thymidine substituted;
- the spin–orbit coupling matrix elements increase with the atomic size of the substituent and are large enough to ensure efficient intersystem crossing;
- the substitution in position 4 is more efficient that that in position 2 of the thymine ring;

Although more detailed information can be obtained employing excited-state simulations, the obtained results clearly emerge that these systems can be considered potential UVA chemotherapeutic agents.

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